Obstructions to deforming curves on a complete intersection of quadrics

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The slides for this lecture is available via this QR code.

Plan of talks

- Motivation (Introduction)
- Main result 1 (Obstructions)
- Main result 2 (Infinitesimal deformations with pole)
- Proof of Main result 1

Motivation: Hilbert schemes of curves

Let X be a projective variety over an algebraically closed field and write

$$\mathbf{Hilb}^{\mathrm{sc}}\, X := \{[C] \in \mathbf{Hilb}\, X \mid C \subset X \text{ smooth and connected}\}.$$

Problem

Construct generically non-reduced components of $\mathbf{Hilb}^{\mathrm{sc}}\, X$ for a given variety X.

The difficulty is to exhibit tangent directions which are not realized by actual deformations.

Known sources of non-reduced components

The problem has a long history.

X	examples and constructions
$\mathbb{P}^3$	Mumford, Gruson–Peskine, Kleppe, Ellia, ...
$\mathbb{P}^n$ $(n \geq 4)$	Ciliberto–Lopez–Miranda, Choi–Iliev–Kim, N., ...
Fano threefolds	Mukai–N., N.

New direction

What can be said for higher-dimensional algebraic varieties?

A result for Fano threefolds

We have obtained the following type of result.

Theorem 1 (Mumford'62 [2],Mukai-N.'09 [1], N.'19 [3])

Let X be a smooth Fano threefold of Picard number one. Then $\mathbf{Hilb}^{\mathrm{sc}}\, X$ contains a generically non-reduced component, whose general member C is contained in a del Pezzo surface or K3 surface

$$S \subset X$$

containing a line or a conic E.

What was known in dimension three

In dimension three, the construction uses surfaces containing a negative curve E.

index	surface S	curve class
≥ 2	del Pezzo surface	$-K_X\|_S + 2E$
1	K3 surface	$-2K_X\|_S + 2E$

Question

Can one find a similar phenomenon for higher-dimensional ambient varieties?

Another motivation: local dimension

There is also a more local question.

Problem

Determine the local dimension of $\mathbf{Hilb}^{sc}\, X$ at a point $[C]$.

This becomes especially subtle when

$$g(C) > 0.$$

In that case, a naive parameter count is often not enough, and one must control all extra tangent directions.

Tool

The Hilbert–flag scheme helps compare deformations of C with deformations of pairs (C, S).

Complete intersections of quadrics

Let

$$S = S_{2,\dots,2} \subset \mathbb{P}^{n+2} = \mathbb{P}$$

be a smooth complete intersection of n quadrics. We study smooth connected curves

$$C \subset S \subset \mathbb{P}.$$

Goal

Understand deformations of C in the ambient projective space $\mathbb{P}$.

Small values of n

The behavior depends strongly on n.

n	$S_{2,\dots,2}$	behavior
1	$S_2 \simeq \mathbb{P}^1 \times \mathbb{P}^1$	curves are unobstructed
2	$S_{2,2} \simeq \mathbf{Bl}_{p_1,\dots,p_5}\, \mathbb{P}^2$	obstructions controlled by $e_1, \dots, e_5$ (N.'26 [5])
≥ 3	K3 or general type	new phenomena

Today

We focus on $n \geq 3$, especially $n = 3$.

The basic setting

Assume

$$n \geq 3, \qquad m \geq 1,$$

and let

$$S = S_{2,\dots,2} \subset \mathbb{P}^{n+2}$$

be smooth. Suppose that S contains a line

$$E \subset S.$$

Put

$$\mathbf{h} := \mathcal{O}_S(1), \qquad D := (n-1)\mathbf{h} + E \in \operatorname{Pic} S.$$

We consider a general curve

$$C \in |mD|.$$

Why the divisor $D = (n-1)\mathrm{h} + E$?

The linear system

$$|D| = |\mathcal{O}_S(n-1)(E)|$$

is related to the contraction of the line E. Since

$$E^2 = 1 - n, \qquad D.E = (n-1)\mathrm{h}.E + E^2 = 0,$$

the morphism defined by $|D|$ contracts E.

Guiding picture

The divisor D is the pull-back of an ample divisor on the contracted surface.

Contraction and vanishing

Let

$$f : S \longrightarrow \overline{S}$$

be the contraction morphism defined by $|D|$. Then there is an ample divisor Δ on $\overline{S}$ such that

$$D = f^*\Delta.$$

Therefore, for $m \gg 0$,

$$H^1(S, O_S(mD)) = 0$$

by Serre vanishing on $\overline{S}$.

Why this matters

The curves $C \in |mD|$ form a well-controlled family, even though $D.E = 0$.

The main theorem: common structure

Let $p = \mathbf{char}\, k \geq 0$.

Theorem 2 (N.'26[4])

Assume that m is sufficiently large and, if $p > 0$, that $p \nmid m$. Let C be a general member of $|mD|$. Then:

1. C is smooth and connected;
2. the Hilbert–flag scheme $\mathbf{HF}\,\mathbb{P}$ is nonsingular at (C, S) of dimension

$$d_{n,m} := \frac{n(n+2)(n+3)}{2} + 2n + 2 + \dim |mD|.$$

Point

The theorem first gives a clean family of pairs $C \subset S \subset \mathbb{P}^{n+2}$ with controlled flag deformations.

Where the dimension $d_{n,m}$ comes from

The number

$$d_{n,m} = \frac{n(n+2)(n+3)}{2} + 2n + 2 + \dim |mD|$$

has a simple meaning.

data	contribution
choice of the line E	$2n+2$
choice of $S_{2,\dots,2}$ containing E	$\frac{n(n+2)(n+3)}{2}$
choice of $C \in \lvert mD \rvert$	$\dim \lvert mD \rvert$

Interpretation

This is the expected local dimension coming from pairs $C \subset S \subset \mathbb{P}^{n+2}$.

The main theorem: the dichotomy

The same construction gives two different outcomes.

K3 case: $n = 3$

The curves C, with E and S varying, form a generically non-reduced component

$$W \subset \mathbf{Hilb}^{sc}\, \mathbb{P}^5.$$

Moreover,

$$h^0(N_{C/\mathbb{P}^5}) = \dim W + 3.$$

Higher-dimensional case: $n > 3$

The corresponding family gives a generically smooth component of dimension $d_{n,m}$.

The headline phenomenon

The theorem says that the complete intersections of quadrics split into two regions:

$n = 3$: generically non-reduced	whereas	$n > 3$: generically smooth

Thus the K3 surface

$$S_{2,2,2} \subset \mathbb{P}^5$$

is exactly the exceptional case in this construction.

Question

What mechanism produces the extra non-reduced directions only in the K3 case?

The K3 case made explicit

For $n = 3$ we have

$$S = S_{2,2,2} \subset \mathbb{P}^5, \qquad D = 2\mathrm{h} + E.$$

For a general curve

$$C \in |m(2\mathrm{h} + E)|,$$

one obtains

$$\deg C = 17m, \qquad g(C) = 17m^2 + 1.$$

The family W has dimension

$$\dim W = d_{3,m} = g(C) + 53.$$

Three extra tangent directions

In the K3 case,

$$h^0(N_{C/\mathbb{P}^5}) = \dim W + 3.$$

Thus there are three tangent directions beyond the directions coming from the family W.

Meaning of the number 3

These three directions are the genuine pole directions along the line E.

The goal is to show that all of them are obstructed.

Why exactly three?

The difference comes from infinitesimal deformations with a pole along E. Since

$$N_{S/\mathbb{P}^5} \simeq \mathcal{O}_S(2)^{\oplus 3},$$

and, for a line $E \subset S_{2,2,2}$,

$$N_{S/\mathbb{P}^5}(E)|_E \simeq \mathcal{O}_E^{\oplus 3},$$

the quotient

$$\frac{H^0(N_{S/\mathbb{P}^5}(E))}{H^0(N_{S/\mathbb{P}^5})}$$

has dimension $\mathbf{3}$.

Question before Theorem 3

Are all genuine pole directions obstructed?

Infinitesimal deformation with a pole

Now assume

$$n = 3, \qquad S = S_{2,2,2} \subset \mathbb{P}^5.$$

Let

$$S^\circ := S \setminus E, \qquad \mathbb{P}^\circ := \mathbb{P}^5 \setminus E.$$

The open immersion $\iota : S^\circ \hookrightarrow S$ gives inclusions

$$N_{S/\mathbb{P}^5}(mE) \hookrightarrow \iota_* N_{S^\circ/\mathbb{P}^\circ}.$$

Thus a section with a pole along E gives a first-order deformation of the open surface $S^\circ \subset \mathbb{P}^\circ$.

The pole-direction quotient

Twisting by E, we have the exact sequence

$$0 \to N_{S/\mathbb{P}^5} \to N_{S/\mathbb{P}^5}(E) \to N_{S/\mathbb{P}^5}(E)|_E \to 0.$$

Since

$$N_{S/\mathbb{P}^5}(E)|_E \simeq \mathcal{O}_E^{\oplus 3},$$

the cokernel of

$$H^0(N_{S/\mathbb{P}^5}) \to H^0(N_{S/\mathbb{P}^5}(E))$$

is three-dimensional.

Interpretation

The quotient records first-order deformations with a genuine simple pole along E.

Theorem 3: every pole direction is obstructed

Theorem 3 (N.'26[4])

Let $S = S_{2,2,2} \subset \mathbb{P}^5$ be a smooth complete intersection containing a line E. For every genuine pole direction

$$\beta \in H^0(N_{S/\mathbb{P}^5}(E)) \setminus H^0(N_{S/\mathbb{P}^5}),$$

the induced first-order deformation β° of $S^\circ \subset \mathbb{P}^\circ$ has nonzero obstruction:

$$\mathbf{ob}(\beta^\circ) \neq 0 \quad \text{in } H^1(N_{S^\circ/\mathbb{P}^\circ}).$$

Geometric meaning of Theorem 3

A section

$$\beta \in H^0(N_{S/\mathbb{P}^5}(E))$$

represents a deformation of S which is allowed to have a simple pole along E.

Statement in words

If the pole is genuine, then the first-order deformation of the open surface $S \setminus E$ cannot be lifted to second order.

Role in Theorem 2

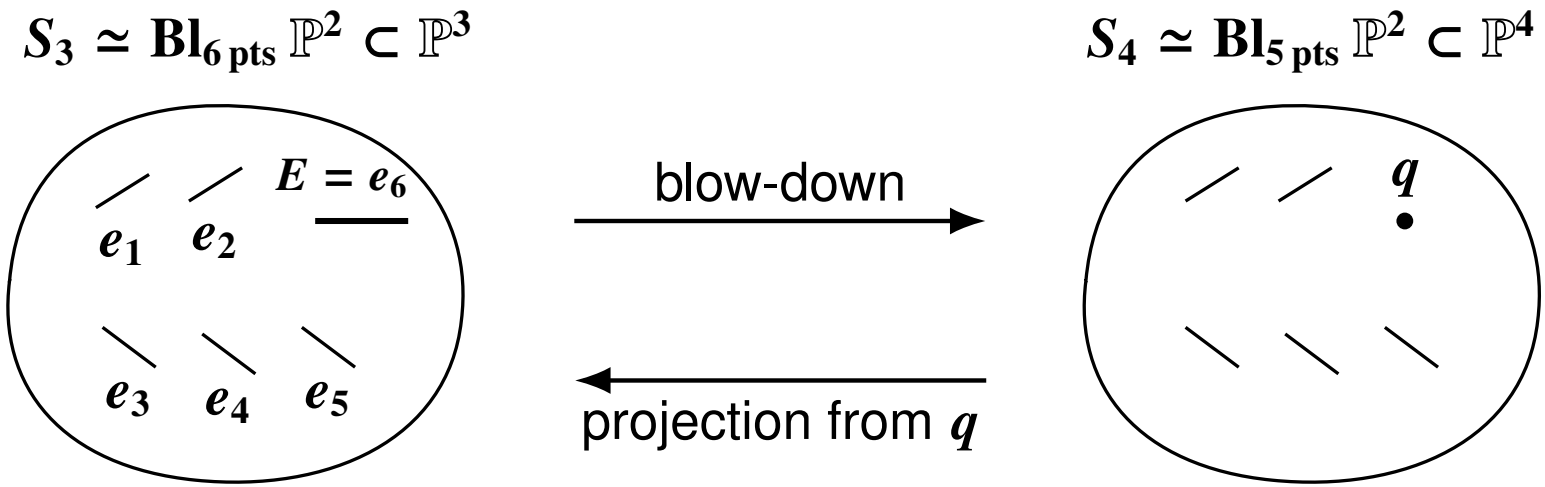
Theorem 3 obstructs all three extra tangent directions in the K3 case.

Example of Unprojection (I)

Here we recall an example of unprojections. A smooth cubic surface S_3 is obtained by blowing up the quartic del Pezzo surface

$$S_4 = (2) \cap (2) \subset \mathbb{P}^4$$

at one point.



Geometric picture

The blow-down contracts the (-1)-curve $E \subset S_3$ to a point $q \in S_4$. Conversely, S_3 is obtained from S_4 by projection from q.

Example of Unprojection (II)

Choose coordinates so that

$$E = (x = y = 0) \subset \mathbb{P}^3.$$

A cubic surface S_3 containing E can be written as

$$S = (Ax + By = 0),$$

where A and B are quadratic forms.

Put

$$f := Ax + By.$$

On S_3, the relation $Ax + By = 0$ gives a global section

$$\theta := \frac{A}{y} = -\frac{B}{x}$$

of $\mathcal{O}_S(1)(E)$.

Example: Unprojection (III)

Lemma 4 (Reid'00 [6])

The blow-down of E from S_3 to S_4 is given by

$$\Phi_{|\mathcal{O}_S(1)(E)|} : S_3 \longrightarrow \mathbb{P}^4, \qquad \mathbf{x} \longmapsto (\mathbf{x}, \theta(x)),$$

which contracts E to a point $q \in S_4$. Moreover, this morphism is the inverse of the projection $\pi_q : S_4 \subset \mathbb{P}^4 \longrightarrow S_3$ from q.

Proof. Introduce a new coordinate t corresponding to θ. Since

$$x\theta = -B, \qquad y\theta = A,$$

the image of S_3 in $\mathbb{P}^4$ is defined by

$$xt + B = 0, \qquad yt - A = 0. \quad \square$$

Interpretation

θ realizes the unprojection $S_3 \to S_4 = (2) \cap (2) \subset \mathbb{P}^4$ of E.

Equations of S containing a line

We go back to our case: $E \subset S_{2,2,2} \subset \mathbb{P}^5$. Choose coordinates so that

$$E = (X_0 = X_1 = X_2 = X_3 = 0) \subset \mathbb{P}^5.$$

A complete intersection of three quadrics containing E can be written as

$$S = (q_1 = q_2 = q_3 = 0),$$

where

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = A \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

for a 3×4 matrix A of linear forms.

The rational section θ

Let A_i be the 3×3 matrix obtained by deleting the i-th column of A, and put

$$\Delta_i = (-1)^i \det A_i.$$

Then on S we have the rational identity

$$\theta := \frac{\Delta_0}{X_0} = \frac{\Delta_1}{X_1} = \frac{\Delta_2}{X_2} = \frac{\Delta_3}{X_3}.$$

Key point

θ is a rational section of $\mathcal{O}_S(2)$, however, it is a global section of $\mathcal{O}_S(2)(E)$.

Computing the obstruction

The explicit description of θ allows one to compute the primary obstruction

$$\mathbf{ob}(\beta) = \beta \cup \mathbf{e} \cup \beta.$$

More precisely, one obtains

$$\mathbf{ob}(\beta) \in H^1(N_{S/\mathbb{P}^5}(3E))$$

and shows that this class is nonzero whenever

$$\beta \notin H^0(N_{S/\mathbb{P}^5}).$$

Message

The pole along E is remembered by the second-order obstruction.

Constructing the family for $n = 3$

Take a line

$$E \subset \mathbb{P}^5, \qquad [E] \in \mathbf{Gr}(2,6).$$

Choose a general complete intersection

$$S = S_{2,2,2} \subset \mathbb{P}^5$$

containing E. Since

$$h^0(\mathbb{P}^5, \mathcal{I}_E(2)) = \binom{7}{2} - h^0(E, \mathcal{O}_E(2)) = 21 - 3 = 18,$$

the pairs (E, S) form an open subset of a Grassmannian bundle over $\mathbf{Gr}(2,6)$.

Dimension of the pair (E, S)

The line E varies in

$$\mathbf{Gr}(2,6), \qquad \dim \mathbf{Gr}(2,6) = 8.$$

For fixed E, the three quadrics defining S span a 3-dimensional subspace of

$$H^0(\mathbb{P}^5, \mathcal{I}_E(2)), \qquad \dim = 18.$$

Hence the fiber has dimension

$$\dim \mathbf{Gr}(3,18) = 3(18-3) = 45.$$

Thus

$$\dim\{(E,S)\} = 8 + 45 = 53.$$

The curves $C \in |m(2\mathrm{h} + E)|$

Now take

$$C \in |m(2\mathrm{h} + E)|$$

general on S. Then

$$\deg C = 17m, \qquad g(C) = 17m^2 + 1.$$

The curves obtained by varying C, E, and S form a family

$$W \subset \mathrm{Hilb}^{\mathrm{sc}}\, \mathbb{P}^5.$$

Since $\dim\{(E, S)\} = 53$, we have

$$\dim W = g(C) + 53 = d_{3,m}.$$

Cohomology behind the tangent excess

The key input is the exact sequence

$$0 \to N_{S/\mathbb{P}^5}(-C) \to N_{S/\mathbb{P}^5}(E-C) \to N_{S/\mathbb{P}^5}(E-C)|_E \to 0.$$

Since $N_{S/\mathbb{P}^5} \simeq \mathcal{O}_S(2H)^{\oplus 3}$ and $C \sim m(2H+E)$, we have

$$N_{S/\mathbb{P}^5}(E-C) \simeq \mathcal{O}_S(-(m-1)D)^{\oplus 3}.$$

For $m \gg 0$, the middle term has no H^0 and no H^1, while

$$N_{S/\mathbb{P}^5}(E-C)|_E \simeq \mathcal{O}_E^{\oplus 3}.$$

Hence

$$H^1(N_{S/\mathbb{P}^5}(-C)) \simeq k^3.$$

Computing tangent directions

For $m \gg 0$, one obtains exact sequences comparing the flag tangent space and the Hilbert tangent space:

$$H^0(N_{(C,S)/\mathbb{P}^5}) \longrightarrow H^0(N_{C/\mathbb{P}^5}) \longrightarrow k^3 \longrightarrow 0.$$

Therefore

$$h^0(N_{C/\mathbb{P}^5}) = \dim W + 3.$$

Interpretation

The three-dimensional quotient is exactly the space of pole directions along E.

Obstructing the three extra directions

Let

$$\alpha \in H^0(N_{C/\mathbb{P}^5})$$

be a tangent direction outside the image of the flag tangent space. Its exterior component lifts to a pole deformation

$$\beta \in H^0(N_{S/\mathbb{P}^5}(E)) \setminus H^0(N_{S/\mathbb{P}^5}).$$

The obstruction theorem for $S \setminus E$ gives

$$\mathbf{ob}(\beta^\circ) \neq 0.$$

By restriction, this implies

$$\mathbf{ob}(\alpha) \neq 0.$$

Conclusion in the K3 case

We have

$$\dim W = d_{3,m}, \qquad h^0(N_{C/\mathbb{P}^5}) = d_{3,m} + 3.$$

The three extra tangent directions are all obstructed.

Conclusion

W is a generically non-reduced component of $\mathbf{Hilb}^{sc}\ \mathbb{P}^5$.

This gives a new family of generically non-reduced components coming from curves on the complete intersection K3 surface $S_{2,2,2} \subset \mathbb{P}^5$.

Summary of talk

1. We studied curves $C \subset S_{2,\ldots,2} \subset \mathbb{P}^{n+2}$.
2. The divisor $D = (n-1)\mathbf{h} + E$ is tied to the contraction of the line E.
3. In the K3 case $S_{2,2,2} \subset \mathbb{P}^5$, pole deformations along E produce three extra tangent directions.
4. These directions have nonzero obstruction.
5. Hence $\mathbf{Hilb}^{\mathrm{sc}}\,\mathbb{P}^5$ has a generically non-reduced component along the family of curves $C \in |m(2\mathbf{h} + E)|$.

Key message

The pole mechanism from Mumford's example persists on complete intersection K3 surfaces.

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